



Yakeen Plus (2025)

Practice Test - 07

DURATION : 200 Minutes

DATE : 06/11/2024

M. MARKS : 720

Topics Covered

Physics :	Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction
Chemistry :	Aldehydes, Ketones and Carboxylic Acids, Amines, Haloalkanes and Haloarenes, Alcohols, Ethers and Phenols
Biology :	(Botany): Photosynthesis in Higher Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plant (Zoology): Animal Kingdom, Biomolecules and Human Reproduction

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are four Section in the Question Paper, Section I, II, III & IV consisting of Section-I (**Physics**), Section-II (**Chemistry**), Section-III (**Botany**) & Section IV (**Zoology**) and having **50 Questions** in each Subject and each subject is divided in two Section, **Section A** consisting 35 questions (all questions are compulsory) and **Section B** consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

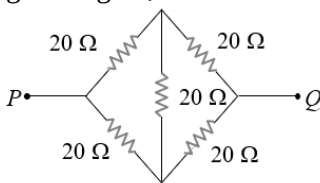
OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

SECTION-(I) PHYSICS

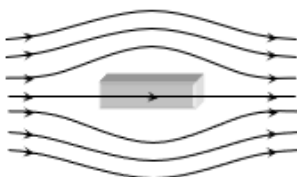
SECTION - A

- The unit of magnetic flux is:
(1) weber/m² (2) weber
(3) henry (4) ampere/m
- An inductor stores energy in:
(1) electric field.
(2) magnetic field.
(3) resistance of coils.
(4) All of the above
- The equivalent resistance between P and Q in the given figure, is:



- (1) 50 Ω (2) 40 Ω
(3) 30 Ω (4) 20 Ω
- Lenz's law is consequence of the law of conservation of:
(1) Charge (2) Momentum
(3) Mass (4) Energy
- A 50 mH coil carries a current of 2 ampere. The energy stored in joules is:
(1) 1 (2) 0.1
(3) 0.05 (4) 0.5
- Kirchhoff's junction law is based on the law of conservation of:
(1) charge.
(2) energy.
(3) momentum.
(4) sum of mass and energy.

- The given figure represents a material which is:



- (1) Paramagnetic. (2) Diamagnetic.
(3) Ferromagnetic. (4) None of these.
- A coil having an area A_0 is placed in a normal magnetic field which changes from B_0 to $4B_0$ in time interval t . The e.m.f. induced in the coil will be:
(1) $3A_0 B_0/t$ (2) $4A_0 B_0/t$
(3) $3B_0/A_0t$ (4) $4A_0/B_0t$

- A piece of copper and another of germanium are cooled from room temperature to 80K. The resistance of:
(1) Each of them increases.
(2) Each of them decreases.
(3) Copper increases and germanium decreases.
(4) Copper decreases and germanium increases.

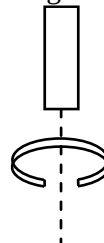
- The relation between magnetic susceptibility χ_m and relative permeability μ_r , is:

- (1) $\chi_m = \mu_r$ (2) $\chi_m - 1 = \mu_r$
(3) $\mu_r = 1 + \chi_m$ (4) $\mu_r = 1 - \chi_m$

- A circular coil of radius 4 cm and of 20 turns carries a current of 3 amperes. It is placed in a magnetic field of intensity of 0.5 weber/m². The magnetic dipole moment of the coil is:

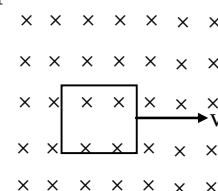
- (1) 0.15 Am² (2) 0.3 Am²
(3) 0.45 Am² (4) 0.6 Am²

- A copper ring having a cut such as not to form a complete loop is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring. Then acceleration of the falling magnet is- (neglect air friction):



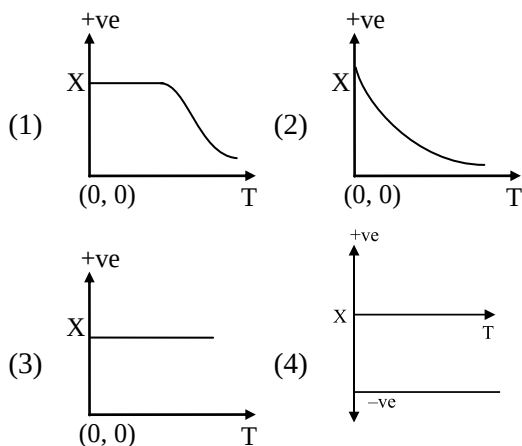
- (1) g (2) less than g
(3) more than g (4) 0

- A square conducting loop of side L and resistance R is moving with a uniform velocity at right angles to one of the sides in its own plane. On applying a uniform magnetic field at right angles to its plane as shown in the figure the induced current in the loop will be:



- (1) Zero
(2) $\frac{BLv}{R}$ in anticlockwise direction
(3) $\frac{BLv}{R}$ in clockwise direction
(4) $\frac{2BLv}{R}$ in clockwise direction

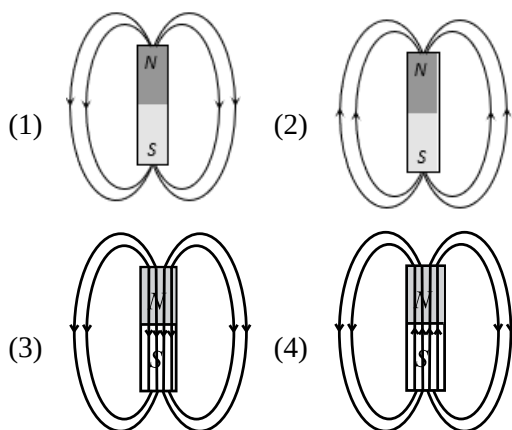
14. The variation of magnetic susceptibility (X) with absolute temperature T for a diamagnetic material is given by:



15. The resistance of a coil is 4.2Ω at 100°C and the temperature coefficient of resistance of its material is $0.004/^\circ \text{C}$. Its resistance at 0°C is:

- (1) 6.5Ω (2) 5Ω
(3) 3Ω (4) 4Ω

16. The magnetic field lines due to a bar magnet are correctly shown in:



17. The maximum power drawn from a cell which is connected across an external variable resistance R , (where r is internal resistance)

- (1) $E^2 / 2r$ (2) $E^2 / 4r$
(3) E^2 / r (4) $E^2 / 3r$

18. The material of permanent magnet has

- (1) High retentivity, low coercivity
(2) Low retentivity, high coercivity
(3) Low retentivity, low coercivity
(4) High retentivity, high coercivity

19. An iron rod of length L and magnetic moment M is bent in the form of a semicircle. Now its magnetic moment will be:

- (1) M
(2) $\frac{2M}{\pi}$
(3) $\frac{M}{\pi}$
(4) $M\pi$

20. In electromagnetic induction, the induced e.m.f. in a coil is independent of:

- (1) Change in the flux
(2) Time
(3) Resistance of the circuit
(4) None of the above

21. Magnetic flux ϕ (in weber) linked with a closed circuit of resistance 10 ohm varies with time t (in seconds) as $\phi = 5t^2 - 4t + 1$. The induced electromotive force in the circuit at $t = 0.2 \text{ sec}$. is:

- (1) 0.4 volts
(2) -0.4 volts
(3) -2.0 volts
(4) 2.0 volts

22. Two identical thin bar magnets, each of length l and pole strength m are placed at right angle to each other, with north pole of one touching south pole of the other. Magnetic moment of the system is:

- (1) ml
(2) $2ml$
(3) $\sqrt{2}ml$
(4) $\frac{1}{2}ml$

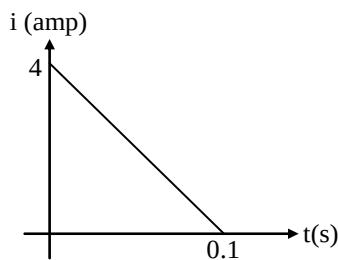
23. A wire of resistance R is stretched till its length is double of the original wire. Then, the resistance of the stretched wire is:

- (1) $2R$
(2) $4R$
(3) $8R$
(4) $16R$

24. The resistance of a wire is 10Ω . Its length is increased by 10% by stretching. The new resistance will now be:

- (1) 12Ω (2) 1.2Ω
(3) 13Ω (4) 11Ω

25. Some magnetic flux is changed from a coil of resistance $10\ \Omega$. As a result, an induced current is developed in it, which varies with time as shown in figure. The magnitude of change in flux through the coil in weber is:



- (1) 2 (2) 4
(3) 6 (4) 8
26. A charged particle enters a magnetic field H with its initial velocity making an angle of 45° with H . The path of the particle will be:
- (1) A straight line
(2) A circle
(3) An ellipse
(4) A helix
27. A magnet of magnetic moment M is situated with its axis along the direction of a magnetic field of strength B . The work done in rotating it by an angle of 180° will be:
- (1) $-MB$
(2) $+MB$
(3) 0
(4) $+2MB$
28. The lowest resistance which can be obtained by connecting 10 resistors each of $1/10\ \Omega$ is:
- (1) $1/250\ \Omega$
(2) $1/200\ \Omega$
(3) $1/100\ \Omega$
(4) $1/10\ \Omega$
29. A galvanometer of $100\ \Omega$ resistance gives full scale deflection when $10\ \text{mA}$ of current is passed. To convert it into $10\ \text{A}$ range ammeter, the resistance of the shunt required will be:

30. In an experiment of meter bridge, a null point is obtained at the centre of the bridge wire, when a resistance of $10\ \Omega$ is connected in one gap. The value of resistance in other gap is:

- (1) $10\ \Omega$
(2) $5\ \Omega$
(3) $\frac{1}{5}\ \Omega$
(4) $500\ \Omega$

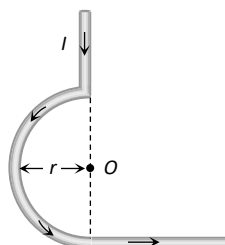
31. The magnetic flux linked with a circuit of resistance $100\ \Omega$ increases from 10 to 60 webers. The amount of induced charge that flows in the circuit is (in coulomb):

- (1) 0.5 (2) 5
(3) 50 (4) 100

32. A coil of N turns and mean cross-sectional area A is rotating with uniform angular velocity ω about an axis at right angle to uniform magnetic field B . The induced e.m.f. E in the coil will be:

- (1) $NBA \sin\omega t$ (2) $NB \omega \sin\omega t$
(3) $NB/A \sin\omega t$ (4) $NBA \omega \sin\omega t$

33. In the figure, what is the magnetic field at the point O :



- (1) $\frac{\mu_0 I}{4\pi r}$
(2) $\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{2\pi r}$
(3) $\frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r}$
(4) $\frac{\mu_0 I}{4r} - \frac{\mu_0 I}{4\pi r}$

34. A long solenoid has n turns per meter and current I A is flowing through it. The magnetic field at the ends of the solenoid is:

- (1) $\frac{\mu_0 n I}{2}$ (2) $\mu_0 n I$
(3) Zero (4) $2\mu_0 n I$

35. The dimension of the magnetic field intensity B is:

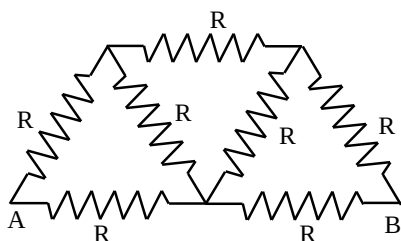
- (1) $MLT^{-2}A^{-1}$
- (2) $MT^{-2}A^{-1}$
- (3) ML^2TA^{-2}
- (4) $M^2LT^{-2}A^{-1}$

SECTION - B

36. An electric field of 1500 V/m and a magnetic field of $0.40 \text{ weber / meter}^2$ act on a moving electron. The minimum uniform speed along a straight line the electron could have is:

- (1) $1.6 \times 10^{15} \text{ m/s}$
- (2) $6 \times 10^{-16} \text{ m/s}$
- (3) $3.75 \times 10^3 \text{ m/s}$
- (4) $3.75 \times 10^2 \text{ m/s}$

37. The equivalent resistance between points A and B in the network shown in fig. is:



- (1) $\frac{15}{7} R$
- (2) $\frac{8}{7} R$
- (3) $\frac{12}{7} R$
- (4) $\frac{16}{7} R$

38. A solenoid of 1.5 metre length and 4.0 cm diameter possess 10 turn per cm . A current of 5 ampere is flowing through it. The magnetic induction at axis inside the solenoid is:

- (1) $2\pi \times 10^{-3} \text{ tesla}$
- (2) $2\pi \times 10^{-5} \text{ tesla}$
- (3) $4\pi \times 10^{-2} \text{ gauss}$
- (4) $2\pi \times 10^{-5} \text{ gauss}$

39. A conductor of 3 m in length is moving perpendicularly to magnetic field of 10^{-3} tesla with the speed of 10^2 m/s , then the e.m.f. produced across the ends of conductor will be:

- (1) 0.03 volt
- (2) 0.3 volt
- (3) $3 \times 10^{-3} \text{ volt}$
- (4) 3 volt

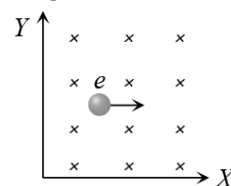
40. 100 mA current gives a full-scale deflection in a galvanometer of 2Ω resistance. The resistance connected with the galvanometer to convert it into a voltmeter to measure 5 V is:

- (1) 98Ω
- (2) 52Ω
- (3) 50Ω
- (4) 48Ω

41. A galvanometer can be converted into an ammeter by connecting:

- (1) Low resistance in series
- (2) High resistance in parallel
- (3) Low resistance in parallel
- (4) High resistance in series

42. In the given figure, the electron enters into the magnetic field. It deflects in direction:

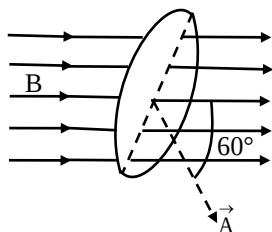


- (1) $+ve X$
- (2) $-ve X$
- (3) $+ve Y$
- (4) $-ve Y$

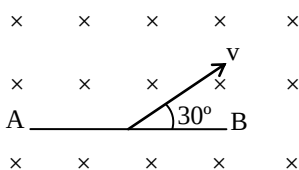
43. A wire of radius r has resistance R . If it is stretched to a radius of $\frac{3r}{4}$, its resistance becomes:

- | | |
|-----------------------|-----------------------|
| (1) $\frac{9R}{16}$ | (2) $\frac{16R}{9}$ |
| (3) $\frac{81R}{256}$ | (4) $\frac{256R}{81}$ |

44. Fig. represents an area $A = 0.5 \text{ m}^2$ situated in a uniform magnetic field $B = 2.0 \text{ weber/m}^2$ and making an angle of 60° with respect to magnetic field. The value of the magnetic flux through the area would be equal to:



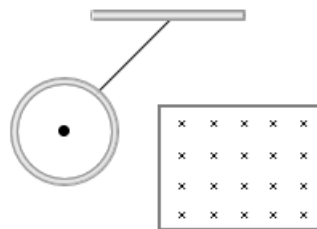
- (1) 2.0 weber
 (2) $\sqrt{3}$ weber
 (3) $\sqrt{3}/2$ weber
 (4) 0.5 weber
45. A conducting rod AB of length $l = 1 \text{ m}$ is moving at a velocity $v = 4 \text{ m/s}$ making an angle 30° with its length. A uniform magnetic field $B = 2 \text{ T}$ exists in a direction perpendicular to the plane of motion. Then:



- (1) $V_A - V_B = 8 \text{ V}$
 (2) $V_A - V_B = 4 \text{ V}$
 (3) $V_B - V_A = 8 \text{ V}$
 (4) $V_B - V_A = 4 \text{ V}$
46. Magnetic intensity for an axial point due to a short bar magnet of magnetic moment M is given by:

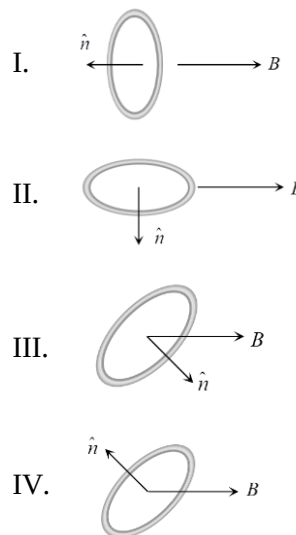
- (1) $\frac{\mu_0}{4\pi} \times \frac{M}{d^3}$
 (2) $\frac{\mu_0}{4\pi} \times \frac{M}{d^2}$
 (3) $\frac{\mu_0}{2\pi} \times \frac{M}{d^3}$
 (4) $\frac{\mu_0}{2\pi} \times \frac{M}{d^2}$

47. A metallic ring connected to a rod oscillates freely like a pendulum. If now a magnetic field is applied in horizontal direction so that the pendulum now swings through the field, the pendulum will:



- (1) Keep oscillating with the old time period
 (2) Keep oscillating with a smaller time period
 (3) Keep oscillating with a larger time period
 (4) Come to rest very soon

48. A current carrying loop is placed in a uniform magnetic field in four different orientations, I, II, III and IV arrange them in the decreasing order of potential energy:



- (1) $I > III > II > IV$
 (2) $I > II > III > IV$
 (3) $I > IV > II > III$
 (4) $III > IV > I > II$

49. Of dia, para and ferromagnetism, the universal property of all substances is:

- (1) diamagnetism.
 (2) paramagnetism.
 (3) ferromagnetism.
 (4) All of the above.

50. The resistivity of a wire depends on its:

- (1) Length.
 (2) Area of cross-section.
 (3) Shape.
 (4) Material.

SECTION-(II) CHEMISTRY**SECTION - A**

51. Which of the following statements is **not** correct?

- (1) Primary amines show intermolecular hydrogen bonding.
- (2) Secondary amines show intermolecular hydrogen bonding.
- (3) Tertiary amines show intermolecular hydrogen bonding.
- (4) Amines have lower boiling points as compared to those of alcohols and carboxylic acids of comparable molar masses.

52. In order to prepare a 1°-Amine from an alkyl halide with simultaneous addition of $-CH_2$ group in the carbon chain, the reagent used as source of nitrogen is:

- (1) Sodium amide, $NaNH_2$
- (2) Sodium azide, NaN_3
- (3) Potassium cyanide, KCN
- (4) Potassium phthalimide, $C_6H_4(CO)_2 N^-K^+$

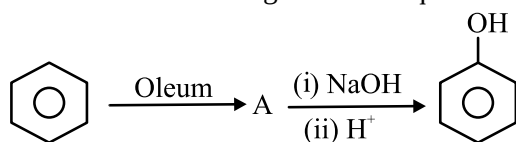
53. The combination which produces tert-Butyl alcohol when treated with Grignard reagent is:

- (1) $CH_3MgBr + CH_3COCH_3$
- (2) $C_2H_5MgBr + CH_3COCH_3$
- (3) $CH_3MgBr + (CH_3)_2CHOH$
- (4) $CH_3MgBr + (CH_3)_3COH$

54. In Reimer-Tiemann reaction dichlorocarbene acts as:

- (1) Nucleophile.
- (2) Electrophile.
- (3) Free radical.
- (4) All of these

55. Consider the following reaction sequence



The compound A is:

- | | |
|-----|-----|
| (1) | (2) |
| (3) | (4) |

56. Which one of the following alcohols cannot be oxidized by K_2CrO_4 ?

- (1) Ethanol
- (2) tert-Butyl alcohol
- (3) Isopropyl alcohol
- (4) Allyl alcohol

57. Which of the following compounds will form a yellow-coloured oily compound on treatment with $NaNO_2$, HCl at $0 - 5^\circ C$ and 2-Naphthol?

- (1)
- (2)
- (3)
- (4)

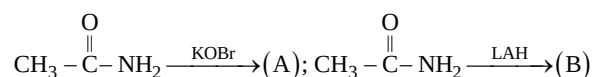
58. In the Hoffmann-Bromamide rearrangement, intermediate compounds are:

- (1) $R - CONHBr$
- (2) $\left[R - \overset{\overset{O}{\parallel}}{C} - \bar{N} - Br \right] Na^+$
- (3) $R - N = C = O$
- (4) All of these

59. High basicity of Me_2NH relative to Me_3N is attributed to:

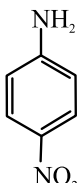
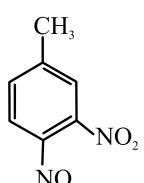
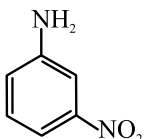
- (1) Effect of solvent.
- (2) Inductive effect of Me.
- (3) Shape of Me_2NH .
- (4) Shape of Me_3N .

60. Relation between (A) and (B) is:

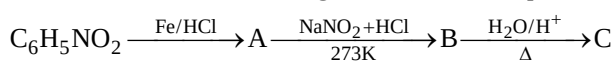


- (1) Chain isomer.
- (2) Homologues.
- (3) Functional isomer.
- (4) Identical isomers.

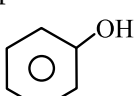
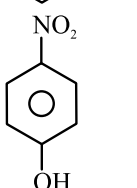
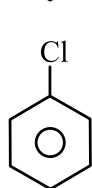
61. An aniline on nitration gives:

- (1)  (2) 
- (3)  (4) Both (1) and (3)

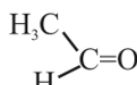
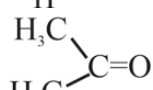
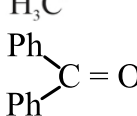
62. Consider the following reaction sequence



The product C is:

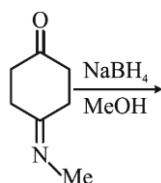
- (1)  (2) 
- (3)  (4) 

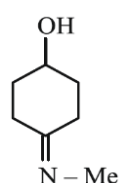
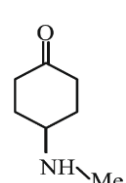
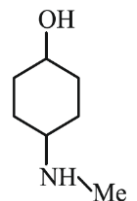
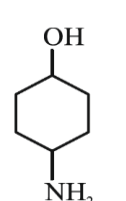
63. The order of reactivity of phenylmagnesium bromide with the following compounds is:

- I. 
- II. 
- III. 

- (1) III > II > I
(2) II > I > III
(3) I > III > II
(4) I > II > III

64. Major product of the following reaction is:



- (1)  (2) 
- (3)  (4) 

65. Glycerol is a:

- (1) Primary alcohol.
(2) Monohydric alcohol.
(3) Secondary alcohol.
(4) Trihydric alcohol.

66. n-Propyl alcohol and Isopropyl alcohol can be chemically distinguished by which reagent?

- (1) PCl_5
(2) Ozonolysis
(3) Oxidation with potassium dichromate
(4) Reduction

67. Phenol is treated with bromine water and shaken well. The white precipitate formed during the process is of:

- (1) 3-Bromophenol
(2) 2,4-Dibromophenol
(3) 2, 4, 6-Tribromophenol
(4) A mixture of 2- and 4-Bromophenol

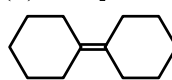
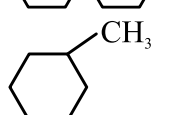
68. Deoxygenation of phenol can be achieved by distillation with:

- (1) Raney nickel.
(2) Lithium aluminium hydride.
(3) Sodium borohydride.
(4) Zinc dust.

69. Cannizzaro's reaction is example of:

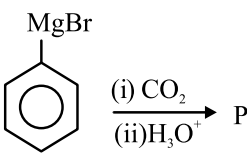
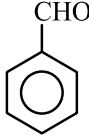
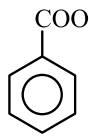
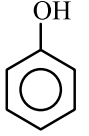
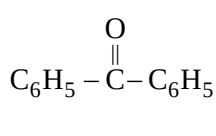
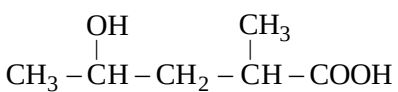
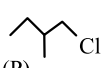
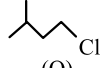
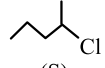
- (1) Redox reaction.
(2) Disproportionation reaction.
(3) Both (1) and (2)
(4) Only oxidation.

70. $\text{X} \xrightarrow[\text{(ii) Zn-H}_2\text{O}]{\text{(i) O}_3} \text{Cyclohexene}=\text{O}$. 'X' can be:

- (1) 
- (2) 
- (3) 
- (4) 

71. What is the product obtained on oxidation of toluene by chromyl chloride followed by hydrolysis?

- (1) Phenol
(2) Benzaldehyde
(3) Benzoic acid
(4) Benzene

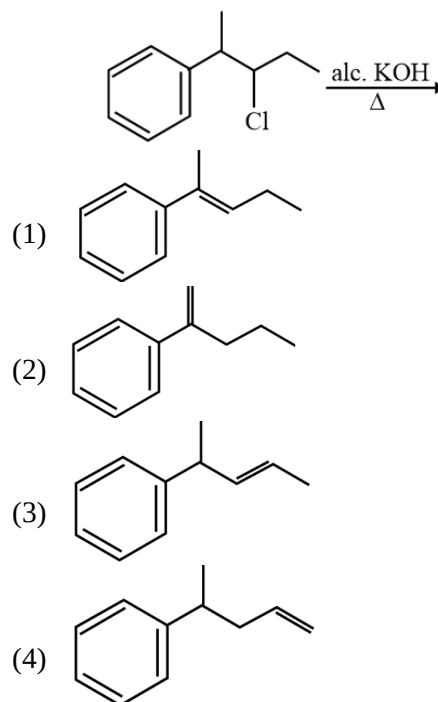
72.  In the above reaction product P is:
- (1)  (2) 
- (3)  (4) 
73. Haloform reaction is **not** given by:
- (1) CH_3COCH_3 (2) $\text{CH}_3\text{COC}_2\text{H}_5$
 (3) $\text{C}_6\text{H}_5\text{COC}_2\text{H}_5$ (4) $\text{CH}_3\text{CHOHCH}_3$
74. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Ketones are less acidic than carboxylic acids.
Reason R: Ketones lacks the acidic hydrogen atom as present in the carboxyl group of carboxylic acids.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.
75. The IUPAC name of the given compound is:
- 
- (1) 1-Hydroxy-1-methylpentanoic acid
 (2) 1-Methyl-3-hydroxypentanoic acid
 (3) 4-Hydroxy-2-methylpentanoic acid
 (4) 4-Carboxypentanol
76. Primary alcohols react with Copper at 573 K to form:
- (1) Aldehyde.
 (2) Ketone.
 (3) Carboxylic acid.
 (4) Alkene.
77. Arrange the following in decreasing order of $\text{S}_{\text{N}}2$ reaction:
- (P)  (Q)  (R)  (S) 

- (1) $\text{P} > \text{Q} > \text{R} > \text{S}$
 (2) $\text{R} > \text{P} > \text{Q} > \text{S}$
 (3) $\text{Q} > \text{R} > \text{P} > \text{S}$
 (4) $\text{R} > \text{Q} > \text{P} > \text{S}$

78. The oxidation of CHCl_3 by air and light is prevented by adding:

- (1) CH_3COOH
 (2) $\text{C}_2\text{H}_5\text{OH}$
 (3) CH_3CHO
 (4) $\text{CH}_3\text{COOCH}_3$

79. Major product of the reaction given below is:



80. $\text{H}_3\text{C}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}=\text{CH}_2 + \text{HBr} \rightarrow (\text{A})$,

(A) predominantly is:

- (1) $\text{CH}_3-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{Br}$
 (2) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3$
 (3) $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{CH}_2\text{Br}$
 (4) $\text{CH}_3-\underset{\text{CH}_3}{\overset{\text{Br}}{\text{C}}}-\text{CH}_2\text{CH}_3$

81. The Reagent used in the gattermann reaction for the preparation of aryl halide is:

- (1) CCl_4
 (2) HBF_4
 (3) Cu/HCl
 (4) HNO_2

82. Alkyl halides are least soluble in water because they:

- (1) Have low boiling points.
- (2) Do not form hydrogen bonding with water.
- (3) Are not miscible with water.
- (4) Are of volatile in nature.

83. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Melting point of 1,4-Dichlorobenzene is more than 1,2-Dichlorobenzene.

Reason R: 1,4-Dichlorobenzene fits in crystal lattice better as compared to 1,2-Dichlorobenzene. In the light of the above statements choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

84. Reaction of ethyl halide with sodium ethoxide is called:

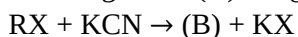
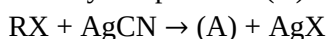
- (1) Wurtz reaction
- (2) Williamson's synthesis
- (3) Elimination reaction
- (4) Carbylamine reaction

85. The sulphonation of chlorobenzene gives exclusively:

- (1) An o-product
- (2) An p-product
- (3) An m-product
- (4) An o-, p-disubstituted product

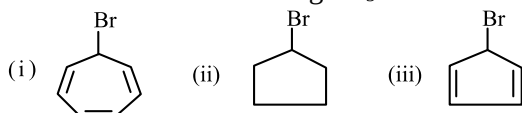
SECTION - B

86. Identify the products (A) and (B) in the reactions:

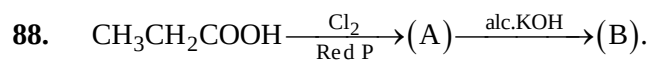


- (1) (A) R – CN, (B) R – CN
- (2) (A) R – CN, (B) R – NC
- (3) (A) R – NC, (B) R – CN
- (4) (A) R – NC, (B) R – NC

87. Decreasing order of reactivity towards S_N1 reaction when treated with $AgNO_3$ is:



- (1) (i) > (iii) > (ii)
- (2) (i) > (ii) > (iii)
- (3) (iii) > (ii) > (i)
- (4) (ii) > (i) > (iii)



The compound (B) is:

- (1) $CH_3 - \underset{\substack{| \\ OH}}{CH} - COOH$
- (2) $\underset{\substack{| \\ OH}}{CH_2}CH_2COOH$
- (3) $H_2C = CHCOOH$
- (4) $ClCH_2CH_2COOH$

89. Which of the following yields a ketone on oxidation?

- (1) $CH_3CH_2CH_2CH_2OH$
- (2) $CH_3CH_2CH(OH)CH_3$
- (3) $(CH_3)_2CHCH_2CH_2OH$
- (4) $CH_3C(CH_3)_2CH_2OH$

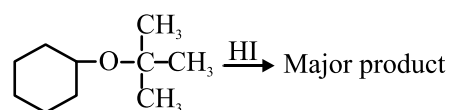
90. C_2H_5CHO and $(CH_3)_2C = O$ can be distinguished by testing with:

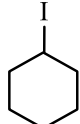
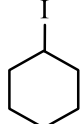
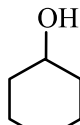
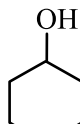
- (1) Phenyl hydrazine
- (2) Hydroxylamine
- (3) Fehling solution
- (4) sodium bisulphite

91. Which of the following does **not** undergo Cannizzaro reaction?

- (1) Benzaldehyde
- (2) 2-Methylpropanal
- (3) 4-Methoxybenzaldehyde
- (4) 2,2-Dimethylpropanal

92. Major product formed in the following reaction is a mixture of:



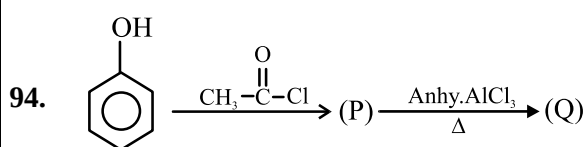
- (1)  and $(CH_3)_3CCl$
- (2)  and $(CH_3)_3COH$
- (3)  and $(CH_3)_3COH$
- (4)  and $CH_3 - \underset{\substack{| \\ CH_3}}{\overset{\substack{CH_3 \\ |}}{C}} - I$

93. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Lucas reagent is a mixture anhydrous ZnCl_2 and concentrated HCl .

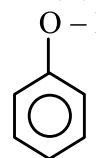
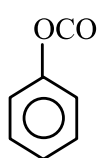
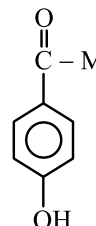
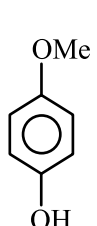
Reason R: Primary alcohol produces precipitate with Lucas reagents.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.



Product (Q) in this reaction is:

- (1) 
- (2) 
- (3) 
- (4) 

95. Why ether is more volatile than an alcohol having same molecular formula?

- (1) Dipolar character of ethers.
- (2) Ethers having resonance structures.
- (3) Inter-molecular hydrogen bonding in ethers.
- (4) Inter-molecular hydrogen bonding in alcohols.

96. A liquid was mixed with ethanol and a drop of concentrated H_2SO_4 was added. A compound with a fruity smell was formed. The liquid was:

- (1) CH_3COCH_3
- (2) CH_3COOH
- (3) CH_3OH
- (4) HCHO

97. Which of the following test can be used to distinguish between ethyl amine and diethyl amine?

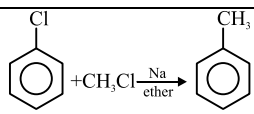
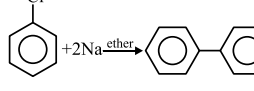
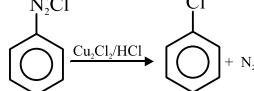
- Carbylamine test
- Nitrous acid test
- Hinsberg's test
- Lucas test
- Ferric chloride test

- (1) Carbylamine test, Nitrous acid test and Ferric chloride test.
- (2) Ferric chloride test, Lucas test and Hinsberg's test.
- (3) Carbylamine test, Nitrous acid test and Hinsberg's test.
- (4) Hinsberg's test, Lucas test and Ferric chloride test.

98. Which is most volatile?

- (1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$
- (2) $(\text{CH}_3)_3\text{N}$
- (3) $\text{CH}_3\text{CH}_2\text{NHCH}_3$
- (4) CH_3OH

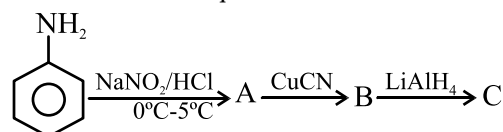
99. Match **List-I** with **List-II**.

List-I		List-II	
(A)		(I)	Fittig reaction
(B)		(II)	Wurtz Fittig reaction
(C)		(III)	Finkelstein reaction
(D)	$\text{C}_2\text{H}_5\text{Cl} + \text{NaI} \xrightarrow{\text{Acetone}} \text{C}_2\text{H}_5\text{I} + \text{NaCl}$	(IV)	Sandmeyer reaction

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-III, D-I
- (4) A-II, B-I, C-IV, D-III

100. In the reaction sequence



The product C is:

- (1) Benzonitrile
- (2) Benzaldehyde
- (3) Benzoic acid
- (4) Benzyl amine

SECTION-(III) BOTANY**SECTION - A**

- 101.** Which of the following statement about sporopollenin is **false**?
- (1) Exine is made up of sporopollenin.
 - (2) Sporopollenin is one of the resistant organic materials.
 - (3) Soluble in acids.
 - (4) Sporopollenin can withstand high temperatures.
- 102.** Select **incorrect** statement regarding microsporogenesis in an anther.
- (1) Inside each microsporangium several thousands of microspores or pollen grains are formed.
 - (2) Each microsporogenesis involves one meiosis and two mitosis.
 - (3) The microspores, as they are formed, are arranged in a cluster of four cells.
 - (4) Cell of the sporogenous tissue is capable of giving rise to a microspore tetrad.
- 103.** In angiosperms each pollen grain produces two male gametes. What do these male gametes do in the fertilisation of a flower?
- (1) Each one fertilises a separate egg cell to give rise to two seeds.
 - (2) One fertilises an egg cell and the other fertilises another cell that gives rise to the tissue of the fruit.
 - (3) Both fertilises a single egg cell.
 - (4) One fertilises an egg cell and the other fertilises the cell that forms food reserve tissue.
- 104.** Which of the following is **not** correct for filiform apparatus?
- (1) It is a finger like projection present towards micropylar end.
 - (2) It guides entry of pollen tube.
 - (3) It is found in the synergid part of embryo sac.
 - (4) It takes part in double fertilisation.

- 105.** Match **List-I** with **List-II**.

List-I		List-II	
(A)	Funicle	(I)	Entry of pollen-tube
(B)	Micropyle of ovule	(II)	Mostly three in number
(C)	Chalazal	(III)	Basal part of ovule
(D)	Antipodals	(IV)	Attached to the placenta

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-I, C-III, D-II
- (3) A-I, B-II, C-III, D-IV
- (4) A-IV, B-III, C-II, D-I

- 106.** As an outbreeding device, self-incompatibility is:
- (1) a genetically controlled mechanism.
 - (2) the germination of pollen grains before its maturity.
 - (3) the germination of pollen grains within the anther.
 - (4) failure of pollination due to lack of pollinating agents.
- 107.** Which of the following sequence shows the **correct** development of female gametophyte in flowering plants?
- (1) Nucellus $\xrightarrow{\text{meiosis}}$ megaspore $\xrightarrow{\text{meiosis}}$ female gametophyte
 - (2) Nucellus $\xrightarrow{\text{mitosis}}$ megaspore mother cell $\xrightarrow{\text{meiosis}}$ megaspore $\xrightarrow{\text{mitosis}}$ female gametophyte
 - (3) Megaspore mother cell $\xrightarrow{\text{meiosis}}$ nucellus $\xrightarrow{\text{mitosis}}$ megaspore $\xrightarrow{\text{meiosis}}$ female gametophyte
 - (4) Nucellus $\xrightarrow{\text{meiosis}}$ megaspore mother cell $\xrightarrow{\text{mitosis}}$ megaspore $\xrightarrow{\text{meiosis}}$ female gametophyte
- 108.** Scutellum, a structure found in embryo of certain plants is a:
- (1) single cotyledons in dicots.
 - (2) single cotyledon in monocots.
 - (3) remains of second cotyledons in monocots.
 - (4) root primordia in dicots.

- 109.** In each microsporangium of every 10 stamens of flowers, there are 20 microspore mother cells. How many pollen grains are formed from that plant?
- (1) 4,000
 - (2) 1,600
 - (3) 16,000
 - (4) 3,200

110. Find the **incorrect** statements w.r.t pollen grain in angiosperms plants.

- A. Germ pores are areas in pollen wall where intine and sporopollenin are absent.
- B. Pollen grains are generally spherical measuring about 25-50 micrometers in diameter.
- C. The variety of architecture seen on the pollen grains.
- D. In over 60 percent of angiosperms, pollen grains are shed at 3-celled stage.

Choose the most appropriate answer from the options given below:

- (1) C and D only (2) A and D only
- (3) B and D only (4) A and C only

111. In an angiosperm, egg apparatus of ovule is situated at _____ **A** _____ end and is composed of _____ **B** _____ cells respectively. Identify A and B.

- | A | B |
|----------------|----------|
| (1) micropylar | two |
| (2) chalazal | three |
| (3) micropylar | three |
| (4) chalazal | two |

112. Perisperm differs from endosperm in:

- (1) having no reserve food.
- (2) being a diploid tissue.
- (3) its formation by fusion of secondary nucleus with several sperms.
- (4) being a haploid tissue.

113. Given below are two statements:

Statement I: Photosynthesis is anabolic, physico-chemical and oxido-reduction process.

Statement II: Photosynthesis occurs in all prokaryotes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

114. O₂ evolved during photosynthesis in green plants comes from water, not from CO₂ was confirmed by:

- (1) experiments of Engelmann using *Cladophora* and aerobic bacteria.
- (2) using radioisotopic techniques.
- (3) half leaf experiments.
- (4) using a suitable H-acceptor molecule like H₂S.

115. RuBisCO enzyme is:

- (1) found only in mesophylls cell of C₄ plants.
- (2) present in the both C₃ and C₄ plants.
- (3) having greater affinity for oxygen when CO₂ concentration is more than O₂.
- (4) the least abundant enzyme on earth.

116. Which of the following event is **not** associated with pollen pistil interaction?

- (1) Pollen deposition on the stigma
- (2) Entry of pollen tube into ovule
- (3) Transfer of pollen grain from anther to the stigma
- (4) Pollen germination on stigma

117. During Electron transport system (ETS) of photosynthesis, H⁺ ions are:

- (1) pumped out from thylakoid lumen always.
- (2) are produced by the splitting of water.
- (3) not required to from ATP.
- (4) always in high concentration in stroma.

118. Read the statements carefully and choose the **correct** option.

- A. Wheat is well adapted to tolerate high temperature.
- B. Bundle sheath cells of maize leaves are larger, thick walled and without intercellular spaces.
- C. Breakdown of C₄ acid of C₄ pathway occurs in bundle sheath cells.
- D. C₄ plants shows double fixation of CO₂.

Choose the most appropriate answer from the options given below:

- (1) A, B and D are correct
- (2) B, C and D are correct
- (3) A and D are incorrect
- (4) B and C are incorrect

119. Choose **incorrect** statement w.r.t mesophyll cells of C₄ plants:

- (1) lack RuBisCO enzyme.
- (2) C₄-acid is broken down to release CO₂ and a 3-carbon molecule.
- (3) primary CO₂ acceptor molecule is PEP.
- (4) have PEPcase enzyme.

120. **Correct** sequence for the Electron transport chain in ETS of thylakoid will be:

- (1) PS-II → Cyt B6f → PQ → PC → PS-I
- (2) PS-I → PC → Cyt B6f → PQ → PS-II
- (3) PS-II → PQ → Cyt B6f → PC → PS-I
- (4) PS-I → Cyt B6f → PC → PQ → PS-II

121. In chloroplasts, the enzyme that catalyse the reduction of NADP^+ to $\text{NADPH} + \text{H}^+$ is located:
- (1) in the lumen of thylakoid.
 - (2) on the inner side of inner membrane.
 - (3) in between the outer and inner membrane.
 - (4) on the stroma side of the thylakoid membrane.

122. Match List-I with List-II.

List-I		List-II	
(A)	Oxaloacetic acid	(I)	3 carbon compound
(B)	Phosphoglyceric acid	(II)	2 carbon compound
(C)	Ribulose 1, 5 bis phosphate	(III)	4 carbon compound
(D)	Phosphoglycolate	(IV)	5 carbon compound

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
 - (2) A-III, B-II, C-IV, D-I
 - (3) A-IV, B-I, C-II, D-III
 - (4) A-III, B-I, C-IV, D-II
123. In Calvin cycle, fixation of 6CO_2 molecules to produce 1 glucose molecule requires:
- (1) 30 ATP and 12 NADPH.
 - (2) 3 ATP and 2 NADPH.
 - (3) 18 ATP and 12 NADPH.
 - (4) 5 ATP and 2 NADPH.
124. Find out the **wrongly** matched pair.
- (1) ABA – Inhibit plant metabolism
 - (2) Cytokinin – Help to produce new leaves
 - (3) Gibberellin – Bolting effect
 - (4) Ethylene – Increases malt yield
125. Plant growth promoters are **not** involved in:
- (1) cell division and cell enlargement.
 - (2) tropic growth and seed formation.
 - (3) dormancy and abscission.
 - (4) fruiting and flowering.
126. Given below are two statements:
- Statement I:** First naturally isolated cytokinin was kinetin from maize.
- Statement II:** Naturally occurring cytokinins, synthesised in root apices.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
 - (2) Statement I is incorrect but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

127. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
- Assertion A:** In C_4 plants photorespiration does not occur.

Reason R: C_4 plants have a mechanism that increases the concentration of CO_2 at the enzyme site.

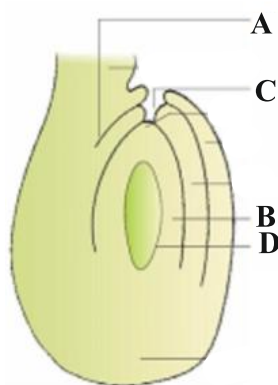
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R are true and R is the correct explanation of A.
 - (4) Both A and R are true but R is NOT the correct explanation of A.
128. Select the **incorrect** option w.r.t ABA.
- (1) It is also called stress hormone.
 - (2) Acts as an antagonist to GAs.
 - (3) Stimulates the opening of stomata.
 - (4) Plays an important role in seed development, maturation and dormancy.
129. Choose **incorrect** option for arithmetic growth.
- (1) Only one daughter cell continues to divide while other differentiates and matures after mitosis.
 - (2) Mathematically expressed as $L_t = L_0 + rt$
 - (3) On plotting the length of the plant organ against time, a S-shaped curve is obtained.
 - (4) It represents constant linear growth.
130. The 'bakanae' disease of rice seedling is caused by a fungal pathogen and the active substance later identified as:
- (1) cytokinin.
 - (2) gibberellic acid.
 - (3) auxin.
 - (4) abscisic acid.
131. The role of auxins in agriculture includes all of the following **except**:
- (1) promoting rooting in stem cuttings.
 - (2) promotes lateral shoot growth and adventitious shoot formation.
 - (3) preventing early fruit and leaf drop.
 - (4) promoting flowering in some plants, like pineapples.
132. Choose **incorrect** option w.r.t ethephon.
- (1) It is an aqueous solution.
 - (2) It is used as source of ethylene.
 - (3) It is used to hasten fruit ripening in tomatoes and apples.
 - (4) It is a natural ripening compound.

133. Identify the phytohormone on the basis of given statements.
- Miller et al. discovered it.
 - Occurs in the region where rapid cell division occurs.
 - Chemically it is modified purine.
 - Help to overcome the apical dominance.
- Choose the most appropriate answer from the options given below:
- Auxin
 - ABA
 - Cytokinin
 - GA
134. The procedure of removal of shoot tip to inhibit apical dominance is:
- capitulation.
 - decapitation.
 - recapitation.
 - bolting.
135. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Cytokinins delay leaf senescence.
Reason R: Cytokinins promote cell division and nutrient mobilisation in plants
 In the light of the above statements, choose the **correct** answer from the options given below:
- A is true but R is false.
 - A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.

SECTION - B

136. Study the figure given below and select the **correct** options.



- | A | B | C | D |
|---------------|----------|-----------|------------|
| (1) Hilum | Nucellus | Micropyle | Embryo sac |
| (2) Micropyle | Hilum | Nucellus | Embryo sac |
| (3) Funicle | Nucellus | Hilum | Embryo sac |
| (4) Hilum | Funicle | Nucellus | Embryo sac |

137. For artificial hybridisation experiment in bisexual flower which of the sequence is **correct**?
- Bagging → Emasculation → Cross pollination → Rebagging
 - Emasculation → Bagging → Cross pollination → Rebagging
 - Cross pollination → Bagging → Emasculation → Rebagging
 - Self-pollination → Bagging → Emasculation → Rebagging
138. Which of the above characters favour entomophily?
- Flowers are usually large, colourful, and fragrant.
 - Large often feathery stigma.
 - Pollen grains are light in weight and non-sticky.
 - Sticky pollen grains.
 - Secrete foul odours.
 - Edible pollen grains and nectar.
- Choose the most appropriate answer from the options given below:
- A, B and D only
 - A, B, C and D only
 - C, D and E only
 - A, D, E and F only
139. Scientists are trying to transfer apomictic genes into hybrid varieties of several of our food and vegetable crops so that:
- they get resistance against common pathogens.
 - the yield of these plants can be increased many folds.
 - hybrid seeds need not to be produced every year.
 - their germplasm is easily stored in labs for further research.
140. The outer layer of the wall of pollen grain:
- is not resistant to enzyme action.
 - exhibits a fascinating array of patterns and design which is of taxonomic significance.
 - has germ pore, where sporopollenin is present.
 - is made up of pectin and hemicellulose.
141. Which of the following events supports ATP synthesis through chemiosmosis?
- Lumen of thylakoid has low electrochemical potential gradient as compared to stroma of chloroplast.
 - Lumen of thylakoid has high H^+ level as compared to stroma of chloroplast.
 - Energy is required to pump protons across a thylakoid membrane.
 - $ATP \rightarrow ADP + P_i$ requires ATP synthase.

142. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Accessory photosynthetic pigments are also called antenna pigments.

Reason R: Antenna pigments absorb light of different wavelengths and transfer the energy to chlorophyll *a* molecule.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

143. Cyclic photophosphorylation differs from non-cyclic photophosphorylation as later involves:

- A. splitting of H_2O .
- B. formation of $NADPH + H^+$.
- C. formation of ATP only.
- D. requirement of external electron source.

Choose the most appropriate answer from the options given below:

- (1) A and B only
- (2) A, B and D only
- (3) B and C only
- (4) C only

144. In some organisms, O_2 is **not** released during photosynthesis. This is because:

- (1) these organisms lack photosynthetic pigments.
- (2) in place of H_2O , these organisms use other molecules as hydrogen donor.
- (3) in such organisms, photosynthetic pigments are sensitive towards O_2 .
- (4) such organisms lack enzymes that dissociate carbon dioxide into its components.

145. Choose the **incorrect** statement.

- (1) In PS I, the reaction centre chlorophyll *a* has an absorption peak at 700 nm.
- (2) The LHC are made up of hundreds of pigment molecules which are bound to proteins.
- (3) Many chlorophyll *a* molecules form the reaction centre.
- (4) Antenna molecules help to make photosynthesis more efficient.

146. Identify the following statements as true (T) or false (F) and choose the option accordingly.

- A. Plant growth is generally indeterminate.
- B. Growth is accompanied by metabolic processes that occur at the expense of energy.
- C. Monocots show open form of growth.
- D. Turgidity of cells helps in extension growth.

(A) (B) (C) (D)

- (1) F F T T
- (2) T F F F
- (3) T T T T
- (4) F T F F

147. Two leaves A and B with surface areas 5 cm^2 and 50 cm^2 respectively, increase their surface areas by 5 cm^2 each in one day.

Select the **correct** statement regarding the growth of these leaves.

- (1) Relative growth rate of leaf A is more than that of leaf B.
- (2) Absolute growth rates of both the leaves are not same.
- (3) Absolute growth rate of leaf A is equal to the relative growth rate of leaf B.
- (4) Growth in leaf B is much faster than that in leaf A.

148. Match List-I with List-II.

List-I (PGRs)		List-II (Chemical composition)	
(A)	Ethylene	(I)	Terpenes
(B)	Absciscic acid	(II)	Gases
(C)	Cytokinin	(III)	Adenine derivatives
(D)	Gibberellins	(IV)	Carotenoid derivatives

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-IV, C-III, D-I
- (3) A-II, B-I, C-III, D-IV
- (4) A-IV, B-III, C-II, D-I

149. Find out the **correct** option w.r.t features of cells of elongation phase.

- (1) Rich in protoplasm.
- (2) Possess large conspicuous nuclei.
- (3) New cell wall deposition.
- (4) Abundant plasmodesmatal connections.

150. The hormone which induces stem elongation in rosette plants also:

- (1) increases the size of the apple.
- (2) induces seed dormancy.
- (3) promotes senescence.
- (4) enhances ripening.

SECTION-(IV) ZOOLOGY

SECTION - A

151. Which of the following statements is **incorrect**?

- (1) The process of fusion of a sperm with an ovum is called fertilisation.
- (2) During fertilisation, a sperm comes in contact with the zona pellucida layer of the ovum.
- (3) The milk produced during the initial few days of lactation is called colostrum.
- (4) Fertilisation takes place in isthmus part of oviduct.

152. Match List-I with List-II.

List-I		List-II	
(A)	Collagen	(I)	Adenosine
(B)	Trypsin	(II)	During breakdown of glucose in anaerobic condition
(C)	Nucleoside	(III)	Intercellular ground substance
(D)	Lactic acid	(IV)	Catalyses breakdown of proteins

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-I, C-IV, D-II

153. Which of the following pairs of nitrogenous bases are pyrimidines?

- (1) Adenine and Guanine
- (2) Cytosine and Thymine
- (3) Thymine and Uracil
- (4) Both (2) and (3)

154. Choose the **incorrect** option.

- (1) *Naja* shed their scales and as skin cast.
- (2) *Aptenodytes* have additional digestive chambers called the crop and gizzard.
- (3) *Ornithorhynchus* is oviparous.
- (4) *Torpedo* possesses poison sting.

155. Given below are two statements.

Statement I: The sperm head contains an elongated haploid nucleus.

Statement II: The mitotic divisions occurring in zygote is called cleavage.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

156. Presence of truly coelomate animals ranges from phylum:

- (1) Aschelminthes to chordata.
- (2) Annelida to chordata.
- (3) Platyhelminthes to chordata.
- (4) Ctenophora to chordata.

157. Read the following statements (A-D).

- A. Cyclostomates are ectoparasite on some fishes.
- B. Air bladder is present in *Pristis* and *Carcharodon*.
- C. Aschelminthes include parasitic as well as non-parasitic roundworms.
- D. Arthropods are the most abundant group with calcareous exoskeleton.

Which of the following statements are **incorrect**?

- (1) B and C only
- (2) B and D only
- (3) A and D only
- (4) A and B only

158. I. Paired accessory gland.

II. Its secretion helps in lubrication of penis.

III. Its secretion, along with other glands, forms seminal plasma.

The above given statements are **correct** for which of the following part of male reproductive system?

- (1) Bulbourethral gland.
- (2) Prostate gland.
- (3) Seminal vesicle.
- (4) Epididymis.

159. Which of the following statements regarding enzymatic inhibitors is **incorrect**?

- (1) Competitive inhibitors bind to the active site of an enzyme.
- (2) Competitive inhibitor closely resembles the substrate in its molecular structure.
- (3) Competitive inhibitors are often used in the control of bacterial pathogens.
- (4) Competitive inhibitors increase the rate of enzyme-catalysed reactions.

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Spermatogenesis starts at the age of puberty.

Reason R: It is due to significant increase in the secretion of gonadotropin releasing hormone (GnRH).

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R true and R is the correct explanation of A.
- (4) Both A and R true and R is not the correct explanation of A.

161. Which of the following is a **correct** matching pair of body features and the animal possessing it?

- (1) Vertebral column- *Salpa* and *Petromyzon*
- (2) Gnathostomata - *Corvus* and *Myxine*
- (3) Air bladder - *Labeo* and *Pristis*
- (4) Pneumatic bones - *Corvus* and *Columba*

162. Which of the following is **not** a carbohydrate in nature?

- (1) GLUT-IV
- (2) Starch
- (3) Glycogen
- (4) Glucose

163. Why is the egg considered haploid?

- (1) It has two X chromosomes.
- (2) It is formed by meiotic division of a diploid primary oocyte.
- (3) It is produced by mitosis.
- (4) It contains one Y chromosome.

164. Complete the analogy and choose the **correct** option.

Carotenoid: Pigment :: Curcumin : _____

- (1) Drug
- (2) Essential oil
- (3) Lectin
- (4) Alkaloid

165. Given below are two statements.

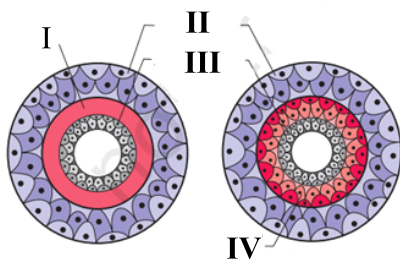
Statement I: In triploblastic animals, a mesoglea layer exists between the ectoderm and endoderm.

Statement II: Proboscis gland is an excretory organ of *Saccoglossus*.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. In the diagram given below Identify parts labelled as I, II, III and IV and select the **incorrect** option.



- (1) I- Mesoderm
- (2) II- Ectoderm
- (3) III- Endoderm
- (4) IV- Mesoderm

167. **Correct** sequence of hormone secretion from beginning of menstruation is:

- (1) FSH, progesterone, estrogen.
- (2) estrogen, FSH, progesterone.
- (3) FSH, estrogen, progesterone.
- (4) estrogen, progesterone, FSH.

168. Choose the **incorrect** match.

(1)	Gametogenesis	Formation of gametes in both males and females.
(2)	Insemination	Transfer of sperms into the female reproductive tract.
(3)	Implantation	Attachment of the blastocyst to the uterine wall.
(4)	Embryogenesis	Formation of the embryo occurs only once in a female's lifetime.

169. Which of the following set of phylums show segmentation?

- (1) Mollusca and Chordata
- (2) Platyhelminthes and Arthropoda
- (3) Echinodermata and Annelida
- (4) Annelida and Arthropoda

170. Read the following statements (A-E).

- A. They have epidermal scales or scutes.
 - B. They have external ear openings.
 - C. Heart is four chambered.
 - D. They are poikilotherms.
 - E. They are oviparous and direct development.
- How many of the above statements is/are **incorrect** w.r.t class reptilia?

- (1) Two
- (2) One
- (3) Four
- (4) Three

171. Which of the following describes the **correct** role of chitin in organisms?

- (1) Chitin serves as a primary energy source in animals.
- (2) Chitin forms the structural component of the animal cell wall.
- (3) Chitin provides structural support in the exoskeleton of arthropods.
- (4) Chitin is the primary storage polysaccharide in fungi.

172. Read the following statements (A-D).

- A. Primary protein provides positional information of amino acids.
- B. Prosthetic groups are organic compounds.
- C. Antibodies fight against infectious agents.
- D. Catalytic activity is lost when the co-factor is removed from the enzyme.

Which of the above statements are **correct**?

- (1) B and C only
- (2) B,C and D only
- (3) A, C and D only
- (4) A, B, C and D

173. Which of the following is **not** the function of placenta?
- (1) Secretes estrogen.
 - (2) Facilitates removal of carbon dioxide and waste material from embryo.
 - (3) Secretes oxytocin during parturition.
 - (4) Facilitates supply of oxygen and nutrients to embryo.

174. The Michaelis constant is a measure of which one of the following?
- (1) Concentration of the enzyme
 - (2) Non-catalytic efficiency of the enzyme
 - (3) Thermo-stability of the enzyme
 - (4) Affinity of the enzyme for its substrate

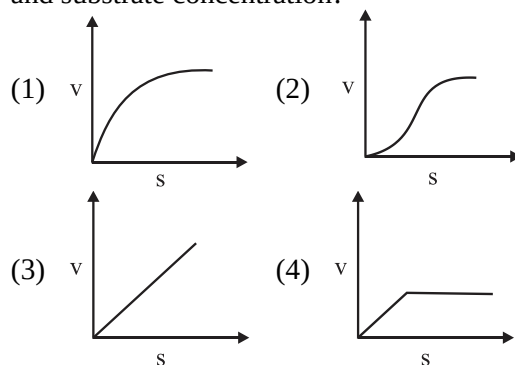
175. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	Parturition	(I)	Present in tertiary follicles
(B)	Antrum	(II)	Cervical canal along with vagina.
(C)	Fimbriae	(III)	Delivery of baby.
(D)	Birth canal	(IV)	Finger like projection in the oviduct.

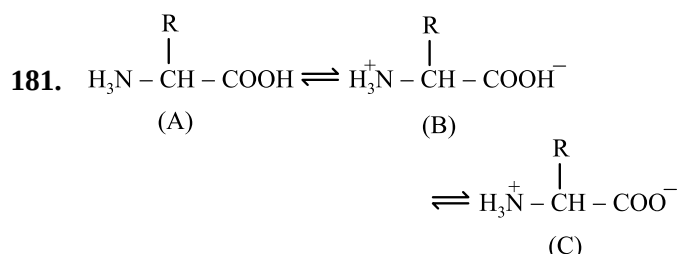
Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
 - (2) A-III, B-I, C-II, D-IV
 - (3) A-III, B-IV, C-I, D-II
 - (4) A-III, B-I, C-IV, D-II
176. Path of sperms from penis to the site of fertilisation is:
- (1) oviduct → uterus → cervix → vagina.
 - (2) vagina → cervix → uterus → oviduct.
 - (3) vagina → uterus → oviduct → cervix.
 - (4) vagina → oviduct → cervix → uterus.
177. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Rate of chemical reaction doubles on decreasing the temperature by 20 degree Celsius.
Reason R: Catalysed reactions proceed at rates vastly higher than that of uncatalysed ones.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R true and R is the correct explanation of A.
 - (4) Both A and R true and R is not the correct explanation of A.
178. Choose the **correctly** matched option.
- (1) Drugs – Abrin
 - (2) Alkaloids – Vinblastin
 - (3) Toxins – Morphine
 - (4) Lectins – Concanavalin A

179. Which of the following graphs best explain the relationship between rate of an enzymatic reaction and substrate concentration?



180. A tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening is:
- (1) hymen.
 - (2) clitoris.
 - (3) seminal vesicle.
 - (4) prostate.



Which of the above is Zwitterionic form?

- (1) A
 - (2) C
 - (3) B
 - (4) All of these
182. Which of the following statement about enzymes is **incorrect**?
- (1) Enzymes are highly specific to their substrates.
 - (2) Enzymes increase the activation energy of biochemical reactions.
 - (3) Most enzymes are proteins.
 - (4) Enzymes can be denatured by extreme temperatures.
183. Choose the **incorrect** match w.r.t % of the total cellular mass.
- | | | |
|-----|---------------|--------|
| (1) | Proteins | 10-15% |
| (2) | Nucleic acids | 5-7% |
| (3) | Carbohydrates | 3% |
| (4) | Lipids | 1% |
184. Given below are two statements.
Statement I: Enzymes are divided into 8 classes each with 4-6 subclasses.
Statement II: Succinate is the substrate for succinic dehydrogenase.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct, but Statement II is incorrect.
 - (2) Statement I is incorrect, but Statement II is correct.
 - (3) Both Statement I and Statement II are correct.
 - (4) Both Statement I and Statement II are incorrect.

185. During parturition, the mild uterine contractions which lead to expulsion of the foetus is known as:
- (1) foetal ejection release.
 - (2) foetal ejection reflex.
 - (3) foetal uterine reflex.
 - (4) foetal placental reflex.

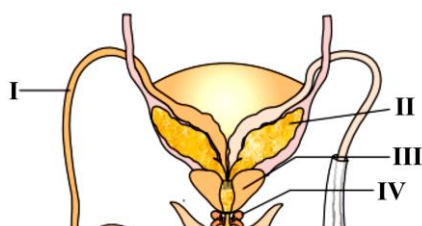
SECTION - B

186. How many primary oocytes are required to produce 50 secondary oocytes?
- (1) 50
 - (2) 100
 - (3) 200
 - (4) 25

187. Which of the following statements about embryonic organ development is **incorrect**?
- (1) After 1 month of pregnancy, the embryo's heart is formed.
 - (2) By the end of 2nd months, embryo are develop the limbs and digits.
 - (3) By the end of 12 weeks, only external genitals are fully formed.
 - (4) During 5th month, the first movements of the foetus can be felt.

188. All the following statements are correct **except**:
- (1) Essential oils are secondary metabolite.
 - (2) Only proteins and lipids are present in the acid insoluble macromolecular fraction.
 - (3) Ctenophora are radial and acoelomate animals.
 - (4) Graffian follicle is a mature follicle of ovary.

189. Identify the any two parts labelled as **I, II, III and IV** and select the **correct** option.



- (1) I-Vas deferens, II-Seminal vesicle
- (2) III-Vas deferens, IV-Bulbourethral gland
- (3) I- Ureter, II-Seminal vesicle
- (4) III-Seminal vesicle, IV-Bulbourethral gland

190. Given below are two statements.

Statement I: In urochordata, the notochord is present only in the larval tail.

Statement II: *Pristis* has the capacity to regulate its body temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct, but Statement II is incorrect.
- (2) Statement I is incorrect, but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

191. Which of the following represents a prosthetic group in a catalase enzyme?

- (1) Coenzyme A
- (2) Zinc ion
- (3) NAD⁺
- (4) Haem

192. Which of the following given statement is **incorrect**?

- (1) Non protein organic part attached loosely to apoenzyme is coenzyme.
- (2) Tyrosine is an aromatic amino acid.
- (3) Palmitic acid has 16 carbons including the carboxyl carbon.
- (4) Adult haemoglobin is an example of quaternary structure.

193. Which of the following given statement is **incorrect**?

- (1) Radial symmetry is exhibited by echinoderms only.
- (2) Sponges are hermaphrodite.
- (3) *Nereis* possess parapodia.
- (4) Radula is a file-like rasping organ for feeding.

194. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Echinus</i>	(I)	Urochordata
(B)	<i>Salpa</i>	(II)	Pelvic fins of male bear claspers.
(C)	<i>Bufo</i>	(III)	Bilaterally symmetrical larvae
(D)	<i>Scoliodon</i>	(IV)	A tympanum represents the ear.

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-I, C-IV, D-II

195. Choose the **incorrect** match.

- (1) Oogonia - Diploid
- (2) Spermatid- Haploid
- (3) Primary spermatocyte- Diploid
- (4) Primary oocyte- Haploid

196. Which of the following is an **incorrect** match?

- (1) *Sycon* – Choanocytes
- (2) *Obelia* – Metagenesis
- (3) *Pleurobrachia* – Bioluminescence
- (4) *Taenia* – Muscular pharynx

197. The starch-I₂ complex is _____ in colour.

Choose the **correct** option to fill the blank.

- (1) yellow. (2) blue.
- (3) green. (4) red.

198. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Bioluminescence is well marked in cnidarians.

Reason R: Bioluminescence is the property of living organism to emit light.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R true and R is the correct explanation of A.
- (4) Both A and R true and R is not the correct explanation of A.

199. Select the **incorrect** statement.

- (1) Protochordates are exclusively marine.
- (2) Ostrich is a flightless reptilian.
- (3) In *Antedon*, excretory system is absent.
- (4) Cyclostomes are ectoparasites on some fishes.

200. Read the following statements (A-E).

- A. The entry of sperm activates the egg for completing mitosis.
- B. Two haploid nuclei fuse to form a diploid zygote.
- C. Relaxin is secreted by the ovary.
- D. Mons pubis is a cushion of fatty tissue covered by pubic hair.
- E. LH acts at the Leydig cells.

How many of the above statements is/are **correct**?

- (1) Two
- (2) One
- (3) Four
- (4) Three

